

STUDY SUMMARY

Final effluent collected from the Cadillac, Michigan Wastewater Treatment Plant (WWTP) was evaluated for the presence of chronic toxicity to larval fathead minnows (*Pimephales promelas*) and *Ceriodaphnia dubia* during August 3-10, 2004. Moderately hard reconstituted water (MHRW) prepared according to United States Environmental Protection Agency (USEPA) guidelines was used as the diluent for the toxicity tests. Since the ammonia level in the original effluent was greater than 5.0 mg/L (6.3 mg/L), the toxicity tests were performed in a CO₂-enriched environment.

The effluent tests met minimum control organism performance requirements defined by the USEPA and the state of Michigan.

The table below summarizes the results of the effluent tests.

Test Endpoint	Fathead Minnow (% Effluent)	<i>Ceriodaphnia dubia</i> (% Effluent)
Acute LC ₅₀ (TU _s)	>100 (0.24)	>100 (0)
Survival NOEC	100	100
Survival LOEC	--	--
Growth/Reproduction NOEC	100	100
Growth/Reproduction LOEC	--	--
Chronic MATC (TU _s)	>100 (0)	>100 (0)

The August 2004 sodium chloride reference toxicity tests for both species met minimum performance requirements defined by the USEPA and the state of Michigan, and produced IC₂₅ values within the expected ranges. Both reference tests indicated the organisms were of acceptable quality and normal sensitivity.

STUDY SUMMARY

Final effluent collected from the Cadillac, Michigan Wastewater Treatment Plant (WWTP) was evaluated for the presence of chronic toxicity to larval fathead minnows (*Pimephales promelas*) and *Ceriodaphnia dubia* during May 10-17, 2005. Moderately hard reconstituted water (MHRW) prepared according to United States Environmental Protection Agency (USEPA) guidelines was used as the diluent for the toxicity tests.

The effluent tests met minimum control organism performance requirements defined by the USEPA and the state of Michigan.

The table below summarizes the results of the effluent tests.

Test Endpoint	Fathead Minnow (% Effluent)	<i>Ceriodaphnia dubia</i> (% Effluent)
Acute LC ₅₀ (TU _s)	>100 (0)	>100 (0)
Survival NOEC	100	100
Survival LOEC	--	--
Growth/Reproduction NOEC	100	100
Growth/Reproduction LOEC	--	--
Chronic MATC (TU _s)	>100 (0)	>100 (0)

The May 2005 sodium chloride reference toxicity tests for both species met minimum performance requirements defined by the USEPA and the state of Michigan, and produced IC₂₅ values within the expected ranges. Both reference tests indicated the organisms were of acceptable quality and normal sensitivity.

STUDY SUMMARY

Final effluent collected from the Cadillac, Michigan Wastewater Treatment Plant (WWTP) was evaluated for the presence of chronic toxicity to larval fathead minnows (*Pimephales promelas*) and *Ceriodaphnia dubia* during March 7-14, 2006. Moderately hard reconstituted water (MHRW) prepared according to United States Environmental Protection Agency (USEPA) guidelines was used as the diluent for the toxicity tests.

The effluent tests met minimum control organism performance requirements defined by the USEPA and the state of Michigan.

The table below summarizes the results of the effluent tests.

Test Endpoint	Fathead Minnow (% Effluent)	<i>Ceriodaphnia dubia</i> (% Effluent)
Acute LC ₅₀ (TU _s)	>100 (0)	>100 (0.2)
Survival NOEC	100	100
Survival LOEC	--	--
Growth/Reproduction NOEC	100	100
Growth/Reproduction LOEC	--	--
Chronic MATC (TU _s)	>100 (0)	>100 (0)

The March 2006 sodium chloride reference toxicity tests for both species met minimum performance requirements defined by the USEPA and the state of Michigan, and produced IC₂₅ values within the expected ranges. Both reference tests indicated the organisms were of acceptable quality and normal sensitivity.



STUDY SUMMARY

Final effluent collected from the Cadillac, Michigan Wastewater Treatment Plant (WWTP) was evaluated for the presence of chronic toxicity to larval fathead minnows (*Pimephales promelas*) and *Ceriodaphnia dubia* during October 10-17, 2006. Moderately hard reconstituted water (MHRW) prepared according to United States Environmental Protection Agency (USEPA) guidelines was used as the diluent for the toxicity tests.

The effluent tests met minimum control organism performance requirements defined by the USEPA and the state of Michigan.

The table below summarizes the results of the effluent tests.

Test Endpoint	Fathead Minnow (% Effluent)	<i>Ceriodaphnia dubia</i> (% Effluent)
Acute LC ₅₀ (TU _s)	>100 (0)	>100 (0)
Survival NOEC	100	100
Survival LOEC	--	--
Growth/Reproduction NOEC	100	100
Growth/Reproduction LOEC	--	--
Chronic MATC (TU _s)	>100 (0)	>100 (0)

The October 2006 sodium chloride reference toxicity tests for both species met minimum performance requirements defined by the USEPA and the state of Michigan, and produced IC₂₅ values within the expected ranges. Both reference tests indicated the organisms were of acceptable quality and normal sensitivity.

KAR Laboratories, Inc.



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City of Cadillac, Water Resources
1121 Plett Rd.
Cadillac, MI 49601

KAR Project No. : 043724

Date Reported : 08/16/04

Date Activated : 08/05/04

Date Due : 08/19/04

Date Validated : 08/13/04

Attn : Mr. Jeff Dietlin

Project

Description : Analysis of one four aqueous samples for ultra-trace mercury.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 043724. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger'.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

KAR Project No. : **043724**

Date Reported : **08/16/04**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of one four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time : **1158**

KAR Sample No. : **043724-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	8/11/2004	PML	
Mercury by EPA 1631	0.6	ng/L	EPA 1631	8/13/2004	PML	

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LABORATORY DETAIL REPORT

KAR Project No. : **043724**

Date Reported : **08/16/04**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of one four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time : **1159**

KAR Sample No. : **043724-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	8/11/2004	PML	
Mercury by EPA 1631	0.6	ng/L	EPA 1631	8/13/2004	PML	

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LABORATORY DETAIL REPORT

KAR Project No. : **043724**

Date Reported : **08/16/04**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of one four aqueous samples for ultra-trace mercury.*

Sample ID : **"Field Blank"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time : **1157**

KAR Sample No. : **043724-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, 1631</i>	<i>Completed</i>		<i>EPA 1631</i>	<i>8/11/2004</i>	<i>PML</i>	
<i>Mercury by EPA 1631</i>	<i><0.5</i>	<i>ng/L</i>	<i>EPA 1631</i>	<i>8/13/2004</i>	<i>PML</i>	

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LABORATORY DETAIL REPORT

KAR Project No. : **043724**

Date Reported : **08/16/04**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of one four aqueous samples for ultra-trace mercury.*

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **043724-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	8/11/2004	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	8/13/2004	PML	

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City of Cadillac, Water Resources
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Cadillac, MI 49601

KAR Project No. : 052286
Date Reported : 06/01/05
Date Activated : 05/20/05
Date Due : 06/06/05
Date Validated : 05/31/05

Attn : Mr. Jeff Dietlin

Project

Description : Analysis of four aqueous samples for ultra-trace mercury.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 052286. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger', is written over a horizontal line.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

KAR Project No. : 052286

Date Reported : 06/01/05

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : 5/20/2005

Sample Date : 5/19/2005

Sample Type : *aqueous*

Sample Time : 1020

KAR Sample No. : 052286-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	5/26/2005	PML	
Mercury by EPA 1631	0.8	ng/L	EPA 1631	5/27/2005	PML	

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LABORATORY DETAIL REPORT

KAR Project No. : **052286**

Date Reported : **06/01/05**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Effluent, Field Duplicate"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **5/20/2005**

Sample Date : **5/19/2005**

Sample Type : **aqueous**

Sample Time : **1020**

KAR Sample No. : **052286-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	5/26/2005	PML	
Mercury by EPA 1631	0.7	ng/L	EPA 1631	5/27/2005	PML	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **052286**

Date Reported : **06/01/05**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Field Blank"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **5/20/2005**

Sample Date : **5/19/2005**

Sample Type : **aqueous**

Sample Time : **1020**

KAR Sample No. : **052286-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	5/26/2005	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	5/27/2005	PML	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **052286**

Date Reported : **06/01/05**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Trip Blank"**

Sampled By :

Sample Date : **5/19/2005**

Sample Time :

Date Received : **5/20/2005**

Sample Type : **aqueous**

KAR Sample No. : **052286-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, 1631</i>	<i>Completed</i>		<i>EPA 1631</i>	<i>5/26/2005</i>	<i>PML</i>	
<i>Mercury by EPA 1631</i>	<i><0.5</i>	<i>ng/L</i>	<i>EPA 1631</i>	<i>5/27/2005</i>	<i>PML</i>	

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**City of Cadillac, Water Resources
1121 Plett Rd.
Cadillac, MI 49601**

**KAR Project No. : 061027
Date Reported : 03/15/06
Date Activated : 03/09/06
Date Due : 03/23/06
Date Validated : 03/15/06**

Attn : Mr. Jeff Dietlin

Project

Description : Analysis of four aqueous samples for ultra-trace mercury.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 061027. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger'.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **061027**

Date Reported : **03/15/06**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : <u>"Tertiary Effluent"</u>						
Sampled By : <i>JMD of City of Cadillac, Water Resources</i>				Date Received : 3/9/2006		
Sample Date : 3/8/2006				Sample Type : aqueous		
Sample Time : 1330				KAR Sample No. : 061027-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, 1631</i>	<i>Completed</i>		<i>EPA 1631</i>	<i>3/13/2006</i>	<i>PML</i>	
<i>Mercury by EPA 1631</i>	<i>0.9</i>	<i>ng/L</i>	<i>EPA 1631</i>	<i>3/14/2006</i>	<i>PML</i>	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **061027**

Date Reported : **03/15/06**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Tertiary Effluent, Field Duplicate"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **3/9/2006**

Sample Date : **3/8/2006**

Sample Type : **aqueous**

Sample Time : **1330**

KAR Sample No. : **061027-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	3/13/2006	PML	
Mercury by EPA 1631	1.2	ng/L	EPA 1631	3/14/2006	PML	

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LABORATORY DETAIL REPORT

KAR Project No. : **061027**

Client: *City of Cadillac, Water Resources*

Date Reported : **03/15/06**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Field Blank"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **3/9/2006**

Sample Date : **3/8/2006**

Sample Type : **aqueous**

Sample Time : **1330**

KAR Sample No. : **061027-03**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	3/13/2006	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	3/14/2006	PML	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **061027**

Date Reported : **03/15/06**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : <u>"Trip Blank"</u>						
Sampled By :				Date Received : 3/9/2006		
Sample Date : 3/8/2006				Sample Type : aqueous		
Sample Time :				KAR Sample No. : 061027-04		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Prep, 1631</i>	<i>Completed</i>		<i>EPA 1631</i>	<i>3/13/2006</i>	<i>PML</i>	
<i>Mercury by EPA 1631</i>	<i><0.5</i>	<i>ng/L</i>	<i>EPA 1631</i>	<i>3/14/2006</i>	<i>PML</i>	

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

KAR Project No. : 064697
Date Reported : 10/26/06
Date Activated : 10/19/06
Date Due : 11/02/06
Date Validated : 10/25/06

Attn : Mr. Jeff Dietlin

Project

Description : Analysis of four aqueous samples for ultra-trace mercury.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 064697. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

KAR Project No. : **064697**

Date Reported : **10/26/06**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : <u>"Tertiary Effluent"</u>						
Sampled By : <i>JMD of City of Cadillac, Water Resources</i>				Date Received : 10/19/2006		
Sample Date : 10/18/2006				Sample Type : aqueous		
Sample Time : 1350				KAR Sample No. : 064697-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	10/23/2006	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	10/24/2006	PML	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **064697**

Date Reported : **10/26/06**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Tertiary Effluent, Field Duplicate"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : **10/19/2006**

Sample Date : **10/18/2006**

Sample Type : **aqueous**

Sample Time : **1350**

KAR Sample No. : **064697-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	10/23/2006	PML	
Mercury by EPA 1631	0.6	ng/L	EPA 1631	10/24/2006	PML	

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LABORATORY DETAIL REPORT

KAR Project No. : 064697

Date Reported : 10/26/06

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Field Blank"**

Sampled By : *JMD of City of Cadillac, Water Resources*

Date Received : 10/19/2006

Sample Date : 10/18/2006

Sample Type : *aqueous*

Sample Time : 1350

KAR Sample No. : 064697-03

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	10/23/2006	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	10/24/2006	PML	

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LABORATORY DETAIL REPORT

KAR Project No. : **064697**

Client: *City of Cadillac, Water Resources*

Date Reported : **10/26/06**

Project

Description : *Analysis of four aqueous samples for ultra-trace mercury.*

Sample ID : **"Trip Blank"**

Sampled By :

Date Received : **10/19/2006**

Sample Date :

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **064697-04**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, 1631	Completed		EPA 1631	10/23/2006	PML	
Mercury by EPA 1631	<0.5	ng/L	EPA 1631	10/24/2006	PML	

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1121 Plett Rd.
Cadillac, MI 49601

Attn : Mr. Jeff Dietlin

KAR Project No. : 043725

Date Reported : 09/20/04

Date Activated : 08/05/04

Date Due : 09/21/04

Date Validated : 09/20/04

Project

Description : Analysis of two aqueous samples for Additional Testing.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 043725. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger'.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **043725**

Date Reported : **09/20/04**

Project

Description : *Analysis of two aqueous samples for Additional Testing.*

Sample ID : **"Composite"**

Sampled By : *RL of City of Cadillac, Water Resources*

Sample Date : **8/4/2004**

Sample Time : **1300**

Date Received : **8/5/2004**

Sample Type : **aqueous**

KAR Sample No. : **043725-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep. metals	Completed		EPA 30xx,200.x	8/6/2004	DMC	
Antimony, total, low level	<2	ug/L	EPA 200.8	8/11/2004	DBL	Elevated detection limit due to sample matrix interference.
Arsenic, total, low level	<1	ug/L	EPA 200.8	8/10/2004	DBL	
Barium, total, low level	22	ug/L	EPA 200.7	9/16/2004	PML	
Beryllium, total, low level	<1	ug/L	EPA 200.8	8/10/2004	DBL	
Boron, total, low level	310	ug/L	EPA 200.7	9/17/2004	PML	
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	8/10/2004	DBL	
Calcium, total	54.4	mg/L	EPA 200.7	8/9/2004	PML	
Chromium, total, low level	<5	ug/L	EPA 200.8	8/10/2004	DBL	
Copper, total, low level	4	ug/L	EPA 200.8	8/10/2004	DBL	
Lead, total, low level	<1	ug/L	EPA 200.8	8/11/2004	DBL	
Magnesium, total	17.0	mg/L	EPA 200.7	8/9/2004	PML	
Nickel, total, low level	5	ug/L	EPA 200.8	8/10/2004	DBL	
Selenium, total, low level	3	ug/L	EPA 200.8	8/11/2004	DBL	
Silver, total, low level	<0.5	ug/L	EPA 200.8	8/11/2004	DBL	
Thallium, total, low level	<2	ug/L	EPA 200.8	8/10/2004	DBL	
Zinc, total, low level	35	ug/L	EPA 200.8	8/10/2004	DBL	
Hardness	206	mg/L (as CaCO3)	SM 2340 B	8/16/2004	BLF	
Prior. Poll. acids	See below		EPA 8270	8/10/2004	KTL	
Prior. Poll. base-neutrals	See below		EPA 8270	8/10/2004	KTL	
Prep. SV Acid/BN	Completed		EPA 3510	8/6/2004	LPV	
1,2,4-Trichlorobenzene 8270	<5	ug/L	EPA 8270	8/10/2004	KTL	
1,2-Dichlorobenzene by 8270	<5	ug/L	EPA 8270	8/10/2004	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 8270	8/10/2004	KTL	
1,3-Dichlorobenzene by 8270	<5	ug/L	EPA 8270	8/10/2004	KTL	
1,4-Dichlorobenzene by 8270	<5	ug/L	EPA 8270	8/10/2004	KTL	
2,3,7,8-TCDD by 8270	<5	ug/L	EPA 8270	8/10/2004	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 8270	8/10/2004	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 8270	8/10/2004	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 8270	8/10/2004	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 8270	8/10/2004	KTL	
2,4-Dinitrotoluene	<5	ug/L	EPA 8270	8/10/2004	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 8270	8/10/2004	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 8270	8/10/2004	KTL	
2-Chlorophenol	<5	ug/L	EPA 8270	8/10/2004	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 8270	8/10/2004	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : **043725**

Client: **City of Cadillac, Water Resources**

Date Reported : **09/20/04**

Project

Description : **Analysis of two aqueous samples for Additional Testing.**

Sample ID : **"Composite"**

Sampled By : **RL of City of Cadillac, Water Resources**

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time : **1300**

KAR Sample No. : **043725-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2-Nitrophenol	<5	ug/L	EPA 8270	8/10/2004	KTL	
3,3'-Dichlorobenzidine	<20	ug/L	EPA 8270	8/10/2004	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 8270	8/10/2004	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 8270	8/10/2004	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 8270	8/10/2004	KTL	
4-Nitrophenol	<20	ug/L	EPA 8270	8/10/2004	KTL	
Acenaphthene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Acenaphthylene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Anthracene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Benidine	<50	ug/L	EPA 8270	8/10/2004	KTL	
Benzo(a)anthracene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Benzo(ghi)perylene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 8270	8/10/2004	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 8270	8/10/2004	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 8270	8/10/2004	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 8270	8/10/2004	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 8270	8/10/2004	KTL	
Chrysene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Di-N-butylphthalate	<5	ug/L	EPA 8270	8/10/2004	KTL	
Di-n-Octyl phthalate	<5	ug/L	EPA 8270	8/10/2004	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Diethyl phthalate	<5	ug/L	EPA 8270	8/10/2004	KTL	
Dimethyl phthalate	<5	ug/L	EPA 8270	8/10/2004	KTL	
Fluoranthene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Fluorene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Hexachlorobenzene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Hexachloroethane	<5	ug/L	EPA 8270	8/10/2004	KTL	
Indeno(123cd)pyrene	<5	ug/L	EPA 8270	8/10/2004	KTL	
Isophorone	<5	ug/L	EPA 8270	8/10/2004	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 8270	8/10/2004	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 8270	8/10/2004	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 8270	8/10/2004	KTL	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **043725**

Date Reported : **09/20/04**

Project

Description : *Analysis of two aqueous samples for Additional Testing.*

Sample ID : **"Composite"**

Sampled By : *RL of City of Cadillac, Water Resources*

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time : **1300**

KAR Sample No. : **043725-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
<i>Naphthalene by Method 8270</i>	<5	ug/L	EPA 8270	8/10/2004	KTL	
<i>Nitrobenzene</i>	<5	ug/L	EPA 8270	8/10/2004	KTL	
<i>Pentachlorophenol</i>	<5	ug/L	EPA 8270	8/10/2004	KTL	
<i>Phenanthrene</i>	<5	ug/L	EPA 8270	8/10/2004	KTL	
<i>Phenol</i>	<5	ug/L	EPA 8270	8/10/2004	KTL	
<i>Pyrene</i>	<5	ug/L	EPA 8270	8/10/2004	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : **043725**

Client: **City of Cadillac, Water Resources**

Date Reported : **09/20/04**

Project

Description : **Analysis of two aqueous samples for Additional Testing.**

Sample ID : **"Grab"**

Sampled By : **RL of City of Cadillac, Water Resources**

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time : **1300**

KAR Sample No. : **043725-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, amenable	<5	ug/L	EPA 335.1	8/10/2004	MTW	Result obtained from total analysis.
Cyanide, total	<5	ug/L	EPA 335.2	8/10/2004	XXX	
Phenols, total	<0.02	mg/L	EPA 420.2	8/9/2004	DMC	
Prior. Poll. volatiles	See below		EPA 8260	8/16/2004	JAR	
Prep. VOA	Completed		EPA 5030	8/16/2004	JAR	
1,1,1-Trichloroethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,1,2-Trichloroethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 8260	8/16/2004	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 8260	8/16/2004	JAR	
2-Chloroethylvinyl ether	<10	ug/L	EPA 8260	8/16/2004	JAR	
Acrolein	<5	ug/L	EPA 8260	8/16/2004	JAR	
Acrylonitrile	<1	ug/L	EPA 8260	8/16/2004	JAR	
Benzene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Bromodichloromethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
Bromoform	<1	ug/L	EPA 8260	8/16/2004	JAR	
Bromomethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
Carbon tetrachloride	<1	ug/L	EPA 8260	8/16/2004	JAR	
Chlorobenzene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Chloroethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
Chloroform	<1	ug/L	EPA 8260	8/16/2004	JAR	
Chloromethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Dibromochloromethane	<1	ug/L	EPA 8260	8/16/2004	JAR	
Ethylbenzene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Methylene chloride	<5	ug/L	EPA 8260	8/16/2004	JAR	
Tetrachloroethene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Toluene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Trichloroethene	<1	ug/L	EPA 8260	8/16/2004	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 8260	8/16/2004	JAR	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **043725**

Date Reported : **09/20/04**

Project

Description : *Analysis of two aqueous samples for Additional Testing.*

Sample ID : **"Grab"**

Sampled By : *RL of City of Cadillac, Water Resources*

Date Received : **8/5/2004**

Sample Date : **8/4/2004**

Sample Type : **aqueous**

Sample Time : **1300**

KAR Sample No. : **043725-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Vinyl chloride	<1	ug/L	EPA 8260	8/16/2004	JAR	

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KAR Laboratories, Inc.



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City of Cadillac, Water Resources
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KAR Project No. : 052136
Date Reported : 05/26/05
Date Activated : 05/12/05
Date Due : 05/26/05
Date Validated : 05/25/05

Attn : Mr. Jeff Dietlin

Project

Description : Analysis of two aqueous samples for NPDES permit.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 052136. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger'.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

KAR Project No. : **052136**

Client: **City of Cadillac, Water Resources**

Date Reported : **05/26/05**

Project

Description : **Analysis of two aqueous samples for NPDES permit.**

Sample ID : **"Plant Effluent, Composite"**

Sampled By : **JDC of City of Cadillac, Water Resources**

Date Received : **5/12/2005**

Sample Date : **5/11/2005**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **052136-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, metals	Completed		EPA 30xx.200.x	5/13/2005	DBL	
Antimony, total, low level	<1	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Arsenic, total, low level	<1	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Barium, total, low level	17	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Beryllium, total, low level	<1	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Boron, total, low level	280	ug/L	EPA 200.7	5/18/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/17/05.
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Calcium, total	60.8	mg/L	EPA 200.7	5/23/2005	DBL	
Chromium, total, low level	<5	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Copper, total, low level	6	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Lead, total, low level	<1	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Magnesium, total	17.1	mg/L	EPA 200.7	5/23/2005	DBL	
Nickel, total, low level	4	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Selenium, total, low level	<1	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Silver, total, low level	<0.5	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Thallium, total, low level	<2	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Zinc, total, low level	64	ug/L	EPA 200.8	5/24/2005	CLB	Measurement was not performed by KAR Labs. Sample digested 5/24/05.
Hardness	222	mg/L (as CaCO3)	SM 2340 B	5/23/2005	BLF	
Prior. Poll. acids	See below		EPA 8270	5/24/2005	KTL	
Prior. Poll. base-neutrals	See below		EPA 8270	5/24/2005	KTL	
Prep, SV Acid/BN	Completed		EPA 3510	5/18/2005	MJY	
1,2,4-Trichlorobenzene 8270	<5	ug/L	EPA 8270	5/24/2005	KTL	
1,2-Dichlorobenzene by 8270	<5	ug/L	EPA 8270	5/24/2005	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 8270	5/24/2005	KTL	
1,3-Dichlorobenzene by 8270	<5	ug/L	EPA 8270	5/24/2005	KTL	
1,4-Dichlorobenzene by 8270	<5	ug/L	EPA 8270	5/24/2005	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 8270	5/24/2005	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : 052136

Client: City of Cadillac, Water Resources

Date Reported : 05/26/05

Project

Description : Analysis of two aqueous samples for NPDES permit.

Sample ID : **"Plant Effluent, Composite"**

Sampled By : JDC of City of Cadillac, Water Resources

Date Received : 5/12/2005

Sample Date : 5/11/2005

Sample Type : aqueous

Sample Time :

KAR Sample No. : 052136-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2,4-Dinitrotoluene	<5	ug/L	EPA 8270	5/24/2005	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 8270	5/24/2005	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 8270	5/24/2005	KTL	
2-Chlorophenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 8270	5/24/2005	KTL	
2-Nitrophenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
3,3'-Dichlorobenzidine	<20	ug/L	EPA 8270	5/24/2005	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 8270	5/24/2005	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 8270	5/24/2005	KTL	
4-Nitrophenol	<20	ug/L	EPA 8270	5/24/2005	KTL	
Acenaphthene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Acenaphthylene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Anthracene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Benzidine	<50	ug/L	EPA 8270	5/24/2005	KTL	
Benzo(a)anthracene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Benzo(ghi)perylene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 8270	5/24/2005	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 8270	5/24/2005	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 8270	5/24/2005	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 8270	5/24/2005	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 8270	5/24/2005	KTL	
Chrysene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Di-N-butylphthalate	<5	ug/L	EPA 8270	5/24/2005	KTL	
Di-n-Octyl phthalate	<5	ug/L	EPA 8270	5/24/2005	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Diethyl phthalate	<5	ug/L	EPA 8270	5/24/2005	KTL	
Dimethyl phthalate	<5	ug/L	EPA 8270	5/24/2005	KTL	
Fluoranthene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Fluorene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Hexachlorobenzene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Hexachloroethane	<5	ug/L	EPA 8270	5/24/2005	KTL	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **052136**

Date Reported : **05/26/05**

Project

Description : *Analysis of two aqueous samples for NPDES permit.*

Sample ID : ***"Plant Effluent, Composite"***

Sampled By : *JDC of City of Cadillac, Water Resources*

Date Received : **5/12/2005**

Sample Date : **5/11/2005**

Sample Type : **aqueous**

Sample Time :

KAR Sample No. : **052136-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Indeno(123cd)pyrene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Isophorone	<5	ug/L	EPA 8270	5/24/2005	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 8270	5/24/2005	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 8270	5/24/2005	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 8270	5/24/2005	KTL	
Naphthalene by Method 8270	<5	ug/L	EPA 8270	5/24/2005	KTL	
Nitrobenzene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Pentachlorophenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
Phenanthrene	<5	ug/L	EPA 8270	5/24/2005	KTL	
Phenol	<5	ug/L	EPA 8270	5/24/2005	KTL	
Pyrene	<5	ug/L	EPA 8270	5/24/2005	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : 052136

Date Reported : 05/26/05

Client: City of Cadillac, Water Resources

Project

Description : Analysis of two aqueous samples for NPDES permit.

Sample ID : **"Plant Effluent, Grab"**

Sampled By : JDC of City of Cadillac, Water Resources

Date Received : 5/12/2005

Sample Date : 5/11/2005

Sample Type : aqueous

Sample Time : 0830

KAR Sample No. : 052136-02

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Phenols, total	<0.02	mg/L	EPA 420.2	5/19/2005	DMC	
Prior. Poll. volatiles	See below		EPA 624	5/20/2005	JAR	
Prep. VOA	Completed		EPA 624	5/20/2005	JAR	
1,1,1-Trichloroethane	<1	ug/L	EPA 624	5/20/2005	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 624	5/20/2005	JAR	
1,1,2-Trichloroethane	<1	ug/L	EPA 624	5/20/2005	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 624	5/20/2005	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 624	5/20/2005	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 624	5/20/2005	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 624	5/20/2005	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 624	5/20/2005	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 624	5/20/2005	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 624	5/20/2005	JAR	
2-Chloroethylvinyl ether	<1	ug/L	EPA 624	5/20/2005	JAR	
Acrolein	<20	ug/L	EPA 624	5/20/2005	JAR	
Acrylonitrile	<2	ug/L	EPA 624	5/20/2005	JAR	
Benzene	<1	ug/L	EPA 624	5/20/2005	JAR	
Bromodichloromethane	<1	ug/L	EPA 624	5/20/2005	JAR	
Bromoform	<1	ug/L	EPA 624	5/20/2005	JAR	
Bromomethane	<1	ug/L	EPA 624	5/20/2005	JAR	
Carbon tetrachloride	<1	ug/L	EPA 624	5/20/2005	JAR	
Chlorobenzene	<1	ug/L	EPA 624	5/20/2005	JAR	
Chloroethane	<1	ug/L	EPA 624	5/20/2005	JAR	
Chloroform	<1	ug/L	EPA 624	5/20/2005	JAR	
Chloromethane	<1	ug/L	EPA 624	5/20/2005	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 624	5/20/2005	JAR	
Dibromochloromethane	<1	ug/L	EPA 624	5/20/2005	JAR	
Ethylbenzene	<1	ug/L	EPA 624	5/20/2005	JAR	
Methylene chloride	<1	ug/L	EPA 624	5/20/2005	JAR	
Tetrachloroethene	<1	ug/L	EPA 624	5/20/2005	JAR	
Toluene	<1	ug/L	EPA 624	5/20/2005	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 624	5/20/2005	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 624	5/20/2005	JAR	
Trichloroethene	<1	ug/L	EPA 624	5/20/2005	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 624	5/20/2005	JAR	
Vinyl chloride	<1	ug/L	EPA 624	5/20/2005	JAR	

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KAR Laboratories, Inc.



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City of Cadillac, Water Resources
1121 Plett Rd.
Cadillac, MI 49601

KAR Project No. : 052237
Date Reported : 05/25/05
Date Activated : 05/18/05
Date Due : 06/02/05
Date Validated : 05/24/05

Attn : Mr. Jeff Dietlin

Project

Description : Analysis of one aqueous sample for available cyanide.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 052237. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger'.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **052237**

Date Reported : **05/25/05**

Project

Description : *Analysis of one aqueous sample for available cyanide.*

Sample ID : **"Tertiary Effluent"**

Sampled By : *SW of City of Cadillac, Water Resources*

Date Received : **5/18/2005**

Sample Date : **5/17/2005**

Sample Type : **aqueous**

Sample Time : **0830**

KAR Sample No. : **052237-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, available	<0.002	mg/L	MCAWW 1677	5/19/2005	CLB	Measurement was not performed by KAR Labs.

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Page 2

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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : Mr. Jeff Dietlin

KAR Project No. : 061023

Date Reported : 03/23/06

Date Activated : 03/09/06

Date Due : 03/23/06

Date Validated : 03/22/06

Project

Description : Analysis of two aqueous samples for NPDES permit.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 061023. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger'.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

KAR Project No. : **061023**

Client: *City of Cadillac, Water Resources*

Date Reported : **03/23/06**

Project

Description : *Analysis of two aqueous samples for NPDES permit.*

Sample ID : **"Plant Effluent, Composite"**

Sampled By :

Date Received : **3/9/2006**

Sample Date : **3/8/2006**

Sample Type : **aqueous**

Sample Time : **0700**

KAR Sample No. : **061023-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, metals	Completed		EPA 30xx,200.x	3/10/2006	DBL	
Antimony, total, low level	<1	ug/L	EPA 200.8	3/16/2006	PML	
Arsenic, total, low level	<1	ug/L	EPA 200.8	3/16/2006	PML	
Barium, total, low level	17	ug/L	EPA 200.8	3/16/2006	PML	
Beryllium, total, low level	<1	ug/L	EPA 200.8	3/16/2006	PML	
Boron, total, low level	390	ug/L	EPA 200.7	3/16/2006	DBL	
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	3/16/2006	PML	
Calcium, total	66.2	mg/L	EPA 200.7	3/17/2006	DBL	
Chromium, total, low level	<5	ug/L	EPA 200.8	3/16/2006	PML	
Copper, total, low level	3	ug/L	EPA 200.8	3/16/2006	PML	
Lead, total, low level	<1	ug/L	EPA 200.8	3/16/2006	PML	
Magnesium, total	21.0	mg/L	EPA 200.7	3/17/2006	DBL	
Nickel, total, low level	4	ug/L	EPA 200.8	3/16/2006	PML	
Selenium, total, low level	<2	ug/L	EPA 200.8	3/17/2006	PML	Elevated detection limit due to sample matrix interference.
Silver, total, low level	<0.5	ug/L	EPA 200.8	3/16/2006	PML	
Thallium, total, low level	<2	ug/L	EPA 200.8	3/16/2006	PML	
Zinc, total, low level	29	ug/L	EPA 200.8	3/16/2006	PML	
Hardness	252	mg/L (as CaCO3)	SM 2340 B	3/17/2006	BLF	
Prior. Poll. acids	See below		EPA 625	3/16/2006	KTL	
Prior. Poll. base-neutrals	See below		EPA 625	3/16/2006	KTL	
Prep, SV Acid/BN	Completed		EPA 625	3/13/2006	MJY	
1,2,4-Trichlorobenzene 8270	<5	ug/L	EPA 625	3/16/2006	KTL	
1,2-Dichlorobenzene by 8270	<5	ug/L	EPA 625	3/16/2006	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 625	3/16/2006	KTL	
1,3-Dichlorobenzene by 8270	<5	ug/L	EPA 625	3/16/2006	KTL	
1,4-Dichlorobenzene by 8270	<5	ug/L	EPA 625	3/16/2006	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 625	3/16/2006	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 625	3/16/2006	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 625	3/16/2006	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 625	3/16/2006	KTL	
2,4-Dinitrotoluene	<5	ug/L	EPA 625	3/16/2006	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 625	3/16/2006	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 625	3/16/2006	KTL	
2-Chlorophenol	<5	ug/L	EPA 625	3/16/2006	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 625	3/16/2006	KTL	
2-Nitrophenol	<5	ug/L	EPA 625	3/16/2006	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : 061023

Client: City of Cadillac, Water Resources

Date Reported : 03/23/06

Project

Description : Analysis of two aqueous samples for NPDES permit.

Sample ID : **"Plant Effluent, Composite"**

Sampled By :

Date Received : 3/9/2006

Sample Date : 3/8/2006

Sample Type : aqueous

Sample Time : 0700

KAR Sample No. : 061023-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
3,3'-Dichlorobenzidine	<20	ug/L	EPA 625	3/16/2006	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 625	3/16/2006	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 625	3/16/2006	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 625	3/16/2006	KTL	
4-Nitrophenol	<20	ug/L	EPA 625	3/16/2006	KTL	
Acenaphthene	<5	ug/L	EPA 625	3/16/2006	KTL	
Acenaphthylene	<5	ug/L	EPA 625	3/16/2006	KTL	
Anthracene	<5	ug/L	EPA 625	3/16/2006	KTL	
Benzidine	<50	ug/L	EPA 625	3/16/2006	KTL	
Benzo(a)anthracene	<5	ug/L	EPA 625	3/16/2006	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 625	3/16/2006	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 625	3/16/2006	KTL	
Benzo(ghi)perylene	<5	ug/L	EPA 625	3/16/2006	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 625	3/16/2006	KTL	
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 625	3/16/2006	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 625	3/16/2006	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 625	3/16/2006	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 625	3/16/2006	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 625	3/16/2006	KTL	
Chrysene	<5	ug/L	EPA 625	3/16/2006	KTL	
Di-N-butylphthalate	<5	ug/L	EPA 625	3/16/2006	KTL	
Di-n-Octyl phthalate	<5	ug/L	EPA 625	3/16/2006	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 625	3/16/2006	KTL	
Diethyl phthalate	<5	ug/L	EPA 625	3/16/2006	KTL	
Dimethyl phthalate	<5	ug/L	EPA 625	3/16/2006	KTL	
Fluoranthene	<5	ug/L	EPA 625	3/16/2006	KTL	
Fluorene	<5	ug/L	EPA 625	3/16/2006	KTL	
Hexachlorobenzene	<5	ug/L	EPA 625	3/16/2006	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 625	3/16/2006	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 625	3/16/2006	KTL	
Hexachloroethane	<5	ug/L	EPA 625	3/16/2006	KTL	
Indeno(123cd)pyrene	<5	ug/L	EPA 625	3/16/2006	KTL	
Isophorone	<5	ug/L	EPA 625	3/16/2006	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 625	3/16/2006	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 625	3/16/2006	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 625	3/16/2006	KTL	
Naphthalene by Method 8270	<5	ug/L	EPA 625	3/16/2006	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : **061023**

Date Reported : **03/23/06**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of two aqueous samples for NPDES permit.*

Sample ID : **"Plant Effluent, Composite"**

Sampled By :

Sample Date : **3/8/2006**

Sample Time : **0700**

Date Received : **3/9/2006**

Sample Type : **aqueous**

KAR Sample No. : **061023-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Nitrobenzene	<5	ug/L	EPA 625	3/16/2006	KTL	
Pentachlorophenol	<5	ug/L	EPA 625	3/16/2006	KTL	
Phenanthrene	<5	ug/L	EPA 625	3/16/2006	KTL	
Phenol	<5	ug/L	EPA 625	3/16/2006	KTL	
Pyrene	<5	ug/L	EPA 625	3/16/2006	KTL	

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Page 4

LABORATORY DETAIL REPORT

KAR Project No. : **061023**

Date Reported : **03/23/06**

Client: **City of Cadillac, Water Resources**

Project

Description : **Analysis of two aqueous samples for NPDES permit.**

Sample ID : **"Plant Effluent, Grab"**

Sampled By :

Date Received : **3/9/2006**

Sample Date : **3/8/2006**

Sample Type : **aqueous**

Sample Time : **0700**

KAR Sample No. : **061023-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, available	<0.002	mg/L	MCAWW 1677	3/14/2006	CLB	Measurement was not performed by KAR Labs.
Phenols, total	<0.02	mg/L	SM 5530 B,D	3/13/2006	EKS	
Prior. Poll. volatiles	See below		EPA 624	3/20/2006	JAR	
Prep. VOA	Completed		EPA 624	3/20/2006	JAR	
1,1,1-Trichloroethane	<1	ug/L	EPA 624	3/20/2006	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 624	3/20/2006	JAR	
1,1,2-Trichloroethane	<1	ug/L	EPA 624	3/20/2006	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 624	3/20/2006	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 624	3/20/2006	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 624	3/20/2006	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 624	3/20/2006	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 624	3/20/2006	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 624	3/20/2006	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 624	3/20/2006	JAR	
2-Chloroethylvinyl ether	<10	ug/L	EPA 624	3/20/2006	JAR	
Acrolein	<20	ug/L	EPA 624	3/20/2006	JAR	
Acrylonitrile	<2	ug/L	EPA 624	3/20/2006	JAR	
Benzene	<1	ug/L	EPA 624	3/20/2006	JAR	
Bromodichloromethane	<1	ug/L	EPA 624	3/20/2006	JAR	
Bromoform	<1	ug/L	EPA 624	3/20/2006	JAR	
Bromomethane	<1	ug/L	EPA 624	3/20/2006	JAR	
Carbon tetrachloride	<1	ug/L	EPA 624	3/20/2006	JAR	
Chlorobenzene	<1	ug/L	EPA 624	3/20/2006	JAR	
Chloroethane	<1	ug/L	EPA 624	3/20/2006	JAR	
Chloroform	<1	ug/L	EPA 624	3/20/2006	JAR	
Chloromethane	<1	ug/L	EPA 624	3/20/2006	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 624	3/20/2006	JAR	
Dibromochloromethane	<1	ug/L	EPA 624	3/20/2006	JAR	
Ethylbenzene	<1	ug/L	EPA 624	3/20/2006	JAR	
Methylene chloride	<1	ug/L	EPA 624	3/20/2006	JAR	
Tetrachloroethene	<1	ug/L	EPA 624	3/20/2006	JAR	
Toluene	<1	ug/L	EPA 624	3/20/2006	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 624	3/20/2006	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 624	3/20/2006	JAR	
Trichloroethene	<1	ug/L	EPA 624	3/20/2006	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 624	3/20/2006	JAR	

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LABORATORY DETAIL REPORT

KAR Project No. : **061023**

Date Reported : **03/23/06**

Client: *City of Cadillac, Water Resources*

Project

Description : *Analysis of two aqueous samples for NPDES permit.*

Sample ID : **"Plant Effluent, Grab"**

Sampled By :

Sample Date : **3/8/2006**

Sample Time : **0700**

Date Received : **3/9/2006**

Sample Type : **aqueous**

KAR Sample No. : **061023-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Vinyl chloride	<1	ug/L	EPA 624	3/20/2006	JAR	

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**City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601**

**KAR Project No. : 064582
Date Reported : 10/26/06
Date Activated : 10/12/06
Date Due : 10/26/06
Date Validated : 10/26/06**

Attn : Mr. Jeff Dietlin

Project

Description : Analysis of two aqueous samples for NPDES permit.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received.

If you wish to contact us about this work please mention KAR Project No. 064582. To arrange additional sampling or testing please contact our Client Services Department. If you have a question regarding quality assurance please contact William Rauch.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Michael J. Jaeger'.

Michael J. Jaeger
Director of Laboratories

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LABORATORY DETAIL REPORT

KAR Project No. : **064582**

Client: **City of Cadillac, Water Resources**

Date Reported : **10/26/06**

Project

Description : Analysis of two aqueous samples for NPDES permit.

Sample ID : "Plant Effluent, Composite"						
Sampled By : RL of City of Cadillac, Water Resources				Date Received : 10/12/2006		
Sample Date : 10/11/2006				Sample Type : aqueous		
Sample Time : 0700				KAR Sample No. : 064582-01		
Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep, metals	Completed		EPA 30xx,200.x	10/13/2006	PML	
Antimony, total, low level	<2	ug/L	EPA 200.8	10/19/2006	PML	Elevated detection limit due to sample matrix interference.
Arsenic, total, low level	<1	ug/L	EPA 200.8	10/19/2006	PML	
Barium, total, low level	25	ug/L	EPA 200.8	10/19/2006	PML	
Beryllium, total, low level	<1	ug/L	EPA 200.8	10/19/2006	PML	
Boron, total, low level	261	ug/L	EPA 200.7	10/24/2006	DBL	
Cadmium, total, low level	<0.2	ug/L	EPA 200.8	10/19/2006	PML	
Calcium, total	59.8	mg/L	EPA 200.7	10/19/2006	DBL	
Chromium, total, low level	<5	ug/L	EPA 200.8	10/19/2006	PML	
Copper, total, low level	3	ug/L	EPA 200.8	10/19/2006	PML	
Lead, total, low level	<1	ug/L	EPA 200.8	10/19/2006	PML	
Magnesium, total	17.1	mg/L	EPA 200.7	10/19/2006	DBL	
Nickel, total, low level	3	ug/L	EPA 200.8	10/19/2006	PML	
Selenium, total, low level	<1	ug/L	EPA 200.8	10/19/2006	PML	
Silver, total, low level	<0.5	ug/L	EPA 200.8	10/19/2006	PML	
Thallium, total, low level	<2	ug/L	EPA 200.8	10/19/2006	PML	
Zinc, total, low level	16	ug/L	EPA 200.8	10/19/2006	PML	
Hardness	220	mg/L (as CaCO3)	SM 2340 B	10/19/2006	BLF	
Prior. Poll. acids	See below		EPA 625	10/19/2006	KTL	
Prior. Poll. base-neutrals	See below		EPA 625	10/19/2006	KTL	
Prep, SV Acid/BN	Completed		EPA 625	10/13/2006	LPV	
1,2,4-Trichlorobenzene 8270	<5	ug/L	EPA 625	10/19/2006	KTL	
1,2-Dichlorobenzene by 8270	<5	ug/L	EPA 625	10/19/2006	KTL	
1,2-Diphenylhydrazine	<5	ug/L	EPA 625	10/19/2006	KTL	
1,3-Dichlorobenzene by 8270	<5	ug/L	EPA 625	10/19/2006	KTL	
1,4-Dichlorobenzene by 8270	<5	ug/L	EPA 625	10/19/2006	KTL	
2,4,6-Trichlorophenol	<5	ug/L	EPA 625	10/19/2006	KTL	
2,4-Dichlorophenol	<5	ug/L	EPA 625	10/19/2006	KTL	
2,4-Dimethylphenol	<5	ug/L	EPA 625	10/19/2006	KTL	
2,4-Dinitrophenol	<20	ug/L	EPA 625	10/19/2006	KTL	
2,4-Dinitrotoluene	<5	ug/L	EPA 625	10/19/2006	KTL	
2,6-Dinitrotoluene	<5	ug/L	EPA 625	10/19/2006	KTL	
2-Chloronaphthalene	<5	ug/L	EPA 625	10/19/2006	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : 064582

Client: City of Cadillac, Water Resources

Date Reported : 10/26/06

Project

Description : Analysis of two aqueous samples for NPDES permit.

Sample ID : **"Plant Effluent, Composite"**

Sampled By : RL of City of Cadillac, Water Resources

Date Received : 10/12/2006

Sample Date : 10/11/2006

Sample Type : aqueous

Sample Time : 0700

KAR Sample No. : 064582-01

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
2-Chlorophenol	<5	ug/L	EPA 625	10/19/2006	KTL	
2-Methyl-4,6-dinitrophenol	<20	ug/L	EPA 625	10/19/2006	KTL	
2-Nitrophenol	<5	ug/L	EPA 625	10/19/2006	KTL	
3,3'-Dichlorobenzidine	<20	ug/L	EPA 625	10/19/2006	KTL	
4-Bromophenyl phenyl ether	<5	ug/L	EPA 625	10/19/2006	KTL	
4-Chloro-3-methylphenol	<5	ug/L	EPA 625	10/19/2006	KTL	
4-Chlorophenyl phenyl ether	<5	ug/L	EPA 625	10/19/2006	KTL	
4-Nitrophenol	<20	ug/L	EPA 625	10/19/2006	KTL	
Acenaphthene	<5	ug/L	EPA 625	10/19/2006	KTL	
Acenaphthylene	<5	ug/L	EPA 625	10/19/2006	KTL	
Anthracene	<5	ug/L	EPA 625	10/19/2006	KTL	
Benzidine	<50	ug/L	EPA 625	10/19/2006	KTL	
Benzo(a)anthracene	<5	ug/L	EPA 625	10/19/2006	KTL	
Benzo(a)pyrene	<5	ug/L	EPA 625	10/19/2006	KTL	
Benzo(b)fluoranthene	<5	ug/L	EPA 625	10/19/2006	KTL	
Benzo(ghi)perylene	<5	ug/L	EPA 625	10/19/2006	KTL	
Benzo(k)fluoranthene	<5	ug/L	EPA 625	10/19/2006	KTL	
Bis(2-chloroethoxy)methane	<5	ug/L	EPA 625	10/19/2006	KTL	
Bis(2-chloroethyl)ether	<5	ug/L	EPA 625	10/19/2006	KTL	
Bis(2-chloroisopropyl)ether	<5	ug/L	EPA 625	10/19/2006	KTL	
Bis(2-ethylhexyl)phthalate	<5	ug/L	EPA 625	10/19/2006	KTL	
Butylbenzyl phthalate	<5	ug/L	EPA 625	10/19/2006	KTL	
Chrysene	<5	ug/L	EPA 625	10/19/2006	KTL	
Di-N-butylphthalate	<5	ug/L	EPA 625	10/19/2006	KTL	
Di-n-Octyl phthalate	<5	ug/L	EPA 625	10/19/2006	KTL	
Dibenzo(ah)anthracene	<5	ug/L	EPA 625	10/19/2006	KTL	
Diethyl phthalate	<5	ug/L	EPA 625	10/19/2006	KTL	
Dimethyl phthalate	<5	ug/L	EPA 625	10/19/2006	KTL	
Fluoranthene	<5	ug/L	EPA 625	10/19/2006	KTL	
Fluorene	<5	ug/L	EPA 625	10/19/2006	KTL	
Hexachlorobenzene	<5	ug/L	EPA 625	10/19/2006	KTL	
Hexachlorobutadiene	<5	ug/L	EPA 625	10/19/2006	KTL	
Hexachlorocyclopentadiene	<5	ug/L	EPA 625	10/19/2006	KTL	
Hexachloroethane	<5	ug/L	EPA 625	10/19/2006	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : **064582**

Client: *City of Cadillac, Water Resources*

Date Reported : **10/26/06**

Project

Description : *Analysis of two aqueous samples for NPDES permit.*

Sample ID : **"Plant Effluent, Composite"**

Sampled By : *RL of City of Cadillac, Water Resources*

Date Received : **10/12/2006**

Sample Date : **10/11/2006**

Sample Type : **aqueous**

Sample Time : **0700**

KAR Sample No. : **064582-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Indeno(123cd)pyrene	<5	ug/L	EPA 625	10/19/2006	KTL	
Isophorone	<5	ug/L	EPA 625	10/19/2006	KTL	
N-Nitrosodi-n-propylamine	<5	ug/L	EPA 625	10/19/2006	KTL	
N-Nitrosodimethylamine	<5	ug/L	EPA 625	10/19/2006	KTL	
N-Nitrosodiphenylamine	<5	ug/L	EPA 625	10/19/2006	KTL	
Naphthalene by Method 8270	<5	ug/L	EPA 625	10/19/2006	KTL	
Nitrobenzene	<5	ug/L	EPA 625	10/19/2006	KTL	
Pentachlorophenol	<5	ug/L	EPA 625	10/19/2006	KTL	
Phenanthrene	<5	ug/L	EPA 625	10/19/2006	KTL	
Phenol	<5	ug/L	EPA 625	10/19/2006	KTL	
Pyrene	<5	ug/L	EPA 625	10/19/2006	KTL	

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LABORATORY DETAIL REPORT

KAR Project No. : 064582

Client: City of Cadillac, Water Resources

Date Reported : 10/26/06

Project

Description : Analysis of two aqueous samples for NPDES permit.

Sample ID : **"Plant Effluent, Grab"**

Sampled By : RL of City of Cadillac, Water Resources

Date Received : 10/12/2006

Sample Date : 10/11/2006

Sample Type : aqueous

Sample Time : 0700

KAR Sample No. : 064582-02

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Cyanide, available	5	ug/L	OIA-1677	10/25/2006	CLB	Measurement was not performed by KAR Labs.
Phenols, total	<0.02	mg/L	SM 5530 B,D	10/20/2006	EKS	
Prior. Poll. volatiles	See below		EPA 624	10/20/2006	JAR	
Prep. VOA	Completed		EPA 624	10/20/2006	JAR	
1,1,1-Trichloroethane	<1	ug/L	EPA 624	10/20/2006	JAR	
1,1,2,2-Tetrachloroethane	<1	ug/L	EPA 624	10/20/2006	JAR	
1,1,2-Trichloroethane	<1	ug/L	EPA 624	10/20/2006	JAR	
1,1-Dichloroethane	<1	ug/L	EPA 624	10/20/2006	JAR	
1,1-Dichloroethene	<1	ug/L	EPA 624	10/20/2006	JAR	
1,2-Dichlorobenzene	<1	ug/L	EPA 624	10/20/2006	JAR	
1,2-Dichloroethane	<1	ug/L	EPA 624	10/20/2006	JAR	
1,2-Dichloropropane	<1	ug/L	EPA 624	10/20/2006	JAR	
1,3-Dichlorobenzene	<1	ug/L	EPA 624	10/20/2006	JAR	
1,4-Dichlorobenzene	<1	ug/L	EPA 624	10/20/2006	JAR	
2-Chloroethylvinyl ether	<10	ug/L	EPA 624	10/20/2006	JAR	
Acrolein	<20	ug/L	EPA 624	10/20/2006	JAR	
Acrylonitrile	<2	ug/L	EPA 624	10/20/2006	JAR	
Benzene	<1	ug/L	EPA 624	10/20/2006	JAR	
Bromodichloromethane	<1	ug/L	EPA 624	10/20/2006	JAR	
Bromoform	<1	ug/L	EPA 624	10/20/2006	JAR	
Bromomethane	<1	ug/L	EPA 624	10/20/2006	JAR	
Carbon tetrachloride	<1	ug/L	EPA 624	10/20/2006	JAR	
Chlorobenzene	<1	ug/L	EPA 624	10/20/2006	JAR	
Chloroethane	<1	ug/L	EPA 624	10/20/2006	JAR	
Chloroform	<1	ug/L	EPA 624	10/20/2006	JAR	
Chloromethane	<1	ug/L	EPA 624	10/20/2006	JAR	
Cis-1,3-Dichloropropene	<1	ug/L	EPA 624	10/20/2006	JAR	
Dibromochloromethane	<1	ug/L	EPA 624	10/20/2006	JAR	
Ethylbenzene	<1	ug/L	EPA 624	10/20/2006	JAR	
Methylene chloride	<1	ug/L	EPA 624	10/20/2006	JAR	
Tetrachloroethene	<1	ug/L	EPA 624	10/20/2006	JAR	
Toluene	<1	ug/L	EPA 624	10/20/2006	JAR	
Trans-1,2-Dichloroethene	<1	ug/L	EPA 624	10/20/2006	JAR	
Trans-1,3-Dichloropropene	<1	ug/L	EPA 624	10/20/2006	JAR	

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **064582**

Date Reported : **10/26/06**

Project

Description : *Analysis of two aqueous samples for NPDES permit.*

Sample ID : **"Plant Effluent, Grab"**

Sampled By : *RL of City of Cadillac, Water Resources*

Date Received : **10/12/2006**

Sample Date : **10/11/2006**

Sample Type : **aqueous**

Sample Time : **0700**

KAR Sample No. : **064582-02**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Trichloroethene	<1	ug/L	EPA 624	10/20/2006	JAR	
Trichlorofluoromethane	<1	ug/L	EPA 624	10/20/2006	JAR	
Vinyl chloride	<1	ug/L	EPA 624	10/20/2006	JAR	

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